

Quantum Phase Detection in the ANNNI Model

To give a quick overview of quantum phase transitions, imagine 8 tiny magnets, or in this case, qubits all in a line. Because of periodic boundary conditions however, this makes these 8 magnets wrap around into a ring, with the eight magnets connecting directly back to the first. When the temperature reaches absolute zero, quantum systems will settle into their lowest possible energy state, also known as its ground state. In this particular 1D Axial Nex-Nearest-Neighbour Ising model, this ground state is essentially a competition between three competing physical forces:

- Neighbor Alignment Force (J_1):** A ferromagnetic coupling that encourages adjacent magnets to point in identical longitudinal directions ($\uparrow\uparrow$).
- Frustration Strength (κ):** An antiferromagnetic next-nearest-neighbor interaction that penalizes parallel alignment between magnets separated by two sites, driving spins two steps apart to anti-align
- External Transverse Field (h):** An external magnetic field applied orthogonally to the spin axis, pulling all magnets sideways into a quantum superposition state ($\rightarrow\rightarrow$).

When the frustration strength κ and transverse field h compete against J_1 , the system experiences **quantum frustration**, which is when there isn't a single classical configuration can simultaneously satisfy every interaction term

Force Name & Symbol	Physical Mechanism	Target Phase Driven
Neighbor Alignment (J_1)	Encourages adjacent spins (i and $i + 1$) to align parallel along the longitudinal Pauli-Z axis ($\uparrow\uparrow$).	Ferromagnetic Phase
Frustration Strength (κ)	Penalizes parallel alignment between next-nearest neighbors (i and $i + 2$), favoring anti-alignment	Antiphase
External Transverse Field (h)	Rotates spins away from the longitudinal axis into a transverse Pauli-X	Paramagnetic Phase

	superposition ($\rightarrow\rightarrow$).	
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Mathematical Formula: 1D ANNNI Hamiltonian

To give a quick overview, a Hamiltonian in the context of quantum mechanics is simply the energy “rulebook” for a quantum system, here is the Hamiltonian for the ANNNI model

$$H = -J \sum_{i=1}^N \sigma_x^i \sigma_x^{i+1} - \kappa \sigma_x^i \sigma_x^{i+2} + h \sigma_z^i,$$

Term-by-Term Hamiltonian Breakdown:

- (Nearest-Neighbor Interaction):** Sets the primary energy scale with $J_1 = 1.0$. The negative prefactor lowers system energy when adjacent spins align in parallel along the Pauli-Z axis, driving ferromagnetic long-range order.
- (Next-Nearest-Neighbor Frustration):** The positive prefactor introduces an energy penalty for parallel next-nearest-neighbor pairs along the Pauli-Z axis. For $\kappa > 0$, this term penalizes alignment and favors anti-aligned pairs , generating quantum frustration against the nearest-neighbor term.
- (Transverse Field Interaction):** Applies a uniform magnetic field along the transverse Pauli-X axis. Increasing h forces spins into transverse superposition states, disrupting longitudinal Z-axis correlations.

Table of Phases

Phase Name	Dominant Parameter Region	Spin Configuration Visual	Physical Meaning
Ferromagnetic	$\kappa < 0.5$, low h	$\uparrow\uparrow\uparrow\uparrow\uparrow\uparrow$	Nearest-neighbor alignment dominates; spins align uniformly along the longitudinal axis.
Antiphase	$\kappa > 0.5$, low h	$\uparrow\uparrow\downarrow\downarrow\uparrow\uparrow\downarrow\downarrow$	Next-nearest-neighbor frustration dominates; spins structure into a repeating 4-step period pattern.

Paramagnetic	High h ($h > h_I, h_{KT}$)	$\rightarrow\rightarrow\rightarrow\rightarrow\rightarrow\rightarrow\rightarrow\rightarrow$	Transverse magnetic field dominates; quantum fluctuations destroy longitudinal Z -correlations.
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To actually detect these magnetic ground states, we can use a variety of different computational methods to get these

Computational Approaches & Methods

To establish an absolute baseline for the 1D ANNNI model, exact ground states are calculated across the 2D parameter plane using classical Exact Diagonalization (ED) with a range of $\kappa \in [0, 1]$, $h \in [0, 2]$. The parameter space will be a 15×15 grid and at each point $2^N \times 2^N$ (256×256 matrix for $N = 8$) Hamiltonian matrix $\hat{H}$ is constructed. We can then solve for the minimum eigenvalue which is the ground state energy and its eigenvector.

Quantum Order Parameters (C1 and C2)

In finite quantum spin systems, single-site magnetization fails as a phase indicator because exact ground states preserve global spin-flip symmetry, yielding across all phases. Two-point spin-correlation functions are therefore required to identify phase boundaries.

Expected Correlation Signatures

- **Ferromagnetic Phase ($\uparrow\uparrow\uparrow$):**
 - $C_1 \approx +1.0$ (Adjacent spins strictly parallel)
 - $C_2 \approx +1.0$ (Spins two sites apart strictly parallel)
- **Antiphase ($\uparrow\uparrow\downarrow$):**
 - $C_1 \approx 0.0$ (Equal balance of parallel $\uparrow\uparrow$ / $\downarrow\downarrow$ and anti-parallel $\uparrow\downarrow$ / $\downarrow\uparrow$ adjacent pairs)
 - $C_2 \approx -1.0$ (Spins two sites apart strictly anti-aligned)
- **Paramagnetic Phase ($\rightarrow\rightarrow\rightarrow\rightarrow$):**
 - $C_1 \approx 0.0$ (Spins aligned along X ; longitudinal Z -correlations vanish)
 - $C_2 \approx 0.0$ (Longitudinal Z -correlations vanish)

Supervised Quantum Machine Learning: QCNN Generalization

Quantum Machine Learning (QML) models process ground states prepared via exact diagonalization or variational quantum eigensolvers (VQE) to construct phase diagrams across discretized grids (15×15 or 20×20).

Supervised Learning: Quantum Convolutional Neural Network (QCNN)

- **Mechanism:** Modeled after classical CNNs, the QCNN applies alternating **convolution layers** (parameterized two-qubit unitaries on adjacent pairs) and **pooling layers** (measuring half the qubits and applying conditional single-qubit rotations to active wires). This iteratively reduces the system dimension until only 2 unmeasured active wires remain.
- **Training:** The network is trained exclusively on simple analytical boundary limits where exact ground state solutions are easy to prepare:
 - $\kappa = 0$ (Transverse-Field Ising limit)
 - $h = 0$ (Quasi-classical Ising limit)

Training minimizes a temperature-sharpened **Cross-Entropy Loss** function over predicted class probabilities

4 Key Physical Insights of Noise

1. **Dampening Effect:** Increasing noise rates ($p = 0.0 \rightarrow 0.01 \rightarrow 0.05$) fade bright correlation colors on heatmaps toward neutral zero (0.0), reducing visual contrast as state purity degrades.
2. **Antiphase Fragility:** The Antiphase relies on delicate next-nearest-neighbor correlations ($C_2 \approx -1.0$). CNOT gate errors quickly disrupt this period-4 ($\uparrow\uparrow\downarrow\downarrow$) structure, causing the Antiphase region to shrink rapidly under noise.
3. **Ferromagnetic Resilience:** Simple uniform spin alignment ($\uparrow\uparrow\uparrow$, where $C_1 \approx +1.0$) retains its characteristic signature significantly better under mild noise ($p = 0.01$) than the Antiphase.
4. **Boundary Smearing:** Sharp theoretical transition boundaries blur into wide, fuzzy transition zones.

Summary & Core Takeaways

1. **Phase Competition:** Quantum phase diagrams emerge directly from the tug-of-war between competing physical interactions: nearest-neighbor alignment (J_1), next-nearest frustration (κ), and external magnetic field forces (h).
2. **Detection Flexibility:** Simple physical correlation order parameters (C_1, C_2) and advanced machine learning models (supervised QCNNs and unsupervised QAD autoencoders) provide complementary, highly accurate pathways to detect quantum phase transitions.
3. **Noise Sensitivity & Mitigation:** Quantum hardware noise scrambles correlation order parameters toward zero, systematically biasing phase diagrams toward the paramagnetic phase. Complex, fragile phases like the Antiphase collapse first under hardware noise, highlighting the critical importance of gate error reduction and error mitigation for near-term quantum processors.